

Osprey Rudder — Engineering Constraints

[T] thesis · [SIM] simulation · [TEAM] stated by the team · **bold** = unverified

Existing rudder. MHZ Mystic C5000, 150 × 30 mm blade, about **3 in (76 mm)** **submerged** on plane (thesis assumed 50%; team agrees; unmeasured). Mount is as low as the transom allows — the previous team had *"concerns that the rudder we had was not long enough to get enough engagement with the water"* [T §3.4.2] . AGFRC A81FHM HV servo, **74 kg·cm stall at 8.4 V**, pull-pull cables [T §3.4.1] .



Installed: transom mount, rudder block atop the blade for the cable actuators. Thesis Fig 5.6.

Constraints

#	CONSTRAINT	REASON
R1	Size on span, not area	Authority is $A_r \cdot CL_a$. Wider chord collapses aspect ratio: 30× the area at fixed span buys 32%; more span buys 5× for a third of the area. [SIM]
R2	Chord stays 30 mm	Follows from R1.
R3	Useful deflection ±10° — limit the autopilot here, not at the stop	Blade ventilates at ~10°; moment then <i>falls</i> (64 → 39 N·m at 30 mph) and stays down until ~6°. [SIM]
R4	Mechanical stop ±35° — assumed, not in thesis	Set by horn and mount tabs. Measure with a protractor.
R5	Stock at 20–25% chord aft of LE	Hinge moment is $F \cdot (x_{cp} - x_{stock}) \cdot c$. Near zero at the CoP; at the LE the lever is 7.5 mm and the servo is overloaded at speed (below).
R6	Mount stays; blade extends downward	Mount cannot go below the transom bottom [T §3.4.2] . Block and cable geometry sit on top and are unchanged.
R7	Same thin section, no end plate	A fat section ventilates earlier; an end plate does nothing once ventilated.
R8	Model valid to 44 mph only	Cavitation number < 0.5 above 19.8 m/s. Numbers past that are extrapolated. [SIM]

Spec — 3 in + 2 in [TEAM]

	CURRENT	NEW		CURRENT	NEW
Submerged span	3.0 in / 76 mm	5.0 in / 127 mm	Total blade, same mount	5.9 in	~8 in
Chord	30 mm	30 mm	Stock position	unknown	20–25% c
Wetted area	3.5 in ²	5.9 in²	Authority $A_r \cdot CL_a$	1.0×	2.2×
Aspect ratio	2.5	4.2	Turn radius, full useful	~20 m (9.5 LOA)	~11.5 m (5.4 LOA)

The course needs < 2° of steady rudder; the gain is margin. Across the parameter uncertainty the current blade sits at its limit 41% of a lap; this one does not saturate [SIM] .

Servo check — 74 kg·cm stall

Peak hinge moment is at ventilation onset (~10°), not full deflection.

SPEED	STOCK AT LE	15% C	20–25% C
30 mph	20 kg·cm (27%)	8 (11%)	~0
45 mph	45 (60%)	18 (24%)	~0
70 mph †	109 (148%)	44 (59%)	~0

† past R8, ±30%. **Sufficient with a balanced stock; overloaded with a leading-edge stock above ~55 mph.** The thesis's 66 kg·cm used the servo horn as the lever; the hinge lever is CoP-to-stock, giving 45 kg·cm worst case at their 80 mph condition [T §3.4.1] [SIM] .

Measure before building: submerged depth on plane (one photo at 30 mph) · current stock position · tiller arm radius · mechanical stop angle · trailer and ramp draft with 2 in more blade.